

VoIP Protocols

Module 45

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0:00 hello viewers I welcome you to the lecture series on cyber security information security today we will

0:07 discuss about V IP protocol that is voice over IP protocol so to start with

0:14 a question arises that what is viip we know that voice can be digitalized the digitalized voice can be transmitted in

0:23 packets over the network Voice transmission comes in the different flavors in the past t communication

0:30 through fixed circuit switch network has dominated in recent years we see data communication Network especially voice

0:39 over IP if you have never heard of VI IP get ready to change the way you think about longdistance phone

0:47 calls viip or voice over Internet Protocol is a method for taking analog

0:54 audio signals like the kind you hear when you talk on the phone and turning them into the Digital Data that can be

1:02 transmitted over the Internet voiceover Internet Protocol is the methodology and group of Technologies for the delivery of voice

1:11 Communications and multimedia sessions over Internet Protocol networks such as Internet Voiceover IP is the

1:20 transmission of voice traffic in packet using the internet as the transmission medium IP is used rather than the

1:29 traditional circuit transmission viip is defined in the

1:34 recommendations it u t h. 32x and RFC

1:42 2443 why viip the Internet Protocol is used to deliver packets carrying digitalized voice however IP was not

1:52 designed for realtime traffic such as voice and video communication IP is the connectionless

1:59 protoc call meaning a virtual connection is not established through a network prior to

2:06 transmission IP makes no guarantees concerning reliability flow control error

detection or error correction

2:14 potential errors include out of sequence packets or even loss of packets Voice transmission requires guaranteed

2:22 connection and a reasonable delay nevertheless IP succeeds partially due to the high cost Associated with the

2:30 traditional circuit switch TDM Network viip uses packet switch Network

2:38 it makes the network transparent to the upper layers that are involved in Voice transmission through an IP based Network

2:47 the existing use of IP network also allows the integration of voice and data integration to leverage the

2:54 connectionless nature of Ip vendors have developed higher layer protocols to address the guaranteed connection and

3:01 3 minutes, 1 second transmission issue viip and OSI model viip follows a layered model comparable

3:09 to the OSI 7 layered model we are familiar with just like the purpose of developing The OSI model breaking into

3:17 defined layers provides a framework it establishes a standard to make the system more manageable and

3:25 flexible each layer is relatively independent to other layers around it changes made to one layer should have no

3:33 or minimal impact on other layers what are the V protocols the two

3:40 most generally used protocols for Vi area unit the itu normal h.323 and also

3:48 the iatf normal sip each area unit signaling protocols that discovered maintain and terminate a

3:58 VI decision Additionally the media entry management protocol mgcp provides a

4:05 signaling and management protocol between viip gateways and ancient pstn public switch telephone Network

4:15 gateways first protocol is itut h.323

4:22 h.323 may be a comprehensive protocol underneath the itut specifications for

4:29 for cation voice video and information across a network the h.323 specification includes

4:39 many subprotocols h225 for specifying decision controls

4:46 example decision setup and tear down h235 for specifying the protection

4:54 framework for h.323 and also the decision setup up

5:02 h245 for specifying media methods and parameter negotiations like terminal

5:10 capabilities .450 for specifying supplementary services like decision hold and

5:17 telephony h235 additionally provides security measures like authentication Integrity

5:25 privacy and a few non repudiation support in h.323

5:32 Communications it's designed to work seamlessly with different protocols like h245 and

5:41 h225 a decision setup is secured through transport layer security TLS once established a decision

5:51 management is initiated so secret writing and media channel data will be

5:57 negotiated h. 32 two three utilizes RTP that is realtime transport protocol or

6:05 rtcp that is realtime transport management protocol as its transport protocol that rides on high of UDP

6:15 secret writing is performed among the RTP packet by thirdparty Hardware or at the network layer authentication

6:23 underneath h.323 will be either radially symmetrical and ention based or subscription

6:32 based for radially symmetrical encryption based authentication previous contact between the ACT entities isn't

6:41 needed as a result of protocol uses defy Helman key exchange to get a

6:48 shared secret identity between the two entities with relevancy the h. 235

6:56 recommendation a subscription based Authentication needs a previous shared secret identity and their area unit

7:04 three variations of this first password based with the radially symmetrical

7:10 secret writing second password based with hashing and third certificate based

7:18 with signatures second protocol is session initiation protocol that is

7:25 sip sip may be a text based application layer protocol that addresses the signaling and session management among

7:33 the packet telephone Network it's outlined in RFC

7:40 3261 sip uses a request response model in sip authentication and authorization

7:48 area unit handled either by employing a lower layer theme or on a request By Request basis with the challenge

7:57 response mechanism as sip may be a lightweight protocol its security capabilities area unit terribly

8:06 restricted sip requests and responses can't be end to end encrypted as a result of message feels like the request

8:14 and root ought to be visible to proxy servers that area unit gift in several Network architectures to make sure sip

8:23 request area unit rooted properly voice information is transmitted in clear text over UDP and

8:32 Communications protocol though sip supports s mbased secret writing exploitation digital certificates sure

8:41 header feels employed in requests and responses can't be encrypted the Sip

protocol depends on

8:48 the transport layer security mechanisms like TLS or IP s to produce the desired security for the entire

8:57 message third Pro protocol is Media entry management protocol that is

9:05 mgcp mgcp is revealed by the media management social unit as RFC

9:13 3435 it expect that mgcp messages can be continuously be carried over Secure net

9:21 connections as outlined within the science security design as outlined in RFC

9:28 2401 exploitation either the science authentication Heather outlined in RFC

9:35 242 or the science encapsulating security payload outlined in RFC

9:42 246 this enables for information connectionless Integrity origin authentication and no obligatory

9:51 anti-replay protection of messages passed between the media entry that converts circuit switch traffic to pack

9:59 based traffic and also the media entry controller that is mgc that dictates the

10:06 service logic of the traffic next is processing in VIP system at the cender aspect analog voice

10:16 signals area unit reborn into digital signals compressed and so set into a planned format exploitation voice codec

10:24 like g.711 g729 and g723

10:32 Etc next the encoded voice is softened into equal Siz packets moreover in every

10:40 packet headers from completely different layers area unit hooked up to the encoded voice the protocol headers

10:48 accessor to voice packets area unit of RTP UDP and signs further as link layer

10:57 header the packet area interet then sent over the network IP to its destination

11:03 wherever the Des forment and deacetalization of the received packets is disbursed throughput the transmission

11:13 method inference time variation of packet delivery might occur hence a playout buffer is employed

11:22 that smoothen the playout at the receiver finish which can have a delay caused as the result of

11:30 interference packets area unit cued at the playout buffer for a stipulated time before being

11:38 y however packets that arrive later than the playout time area unit

11:44 discarded viip uses the sign protocol specifically session initiation protocol

11:51 that is sep and H Point 323 these sign protocols area unit needed to a certain View new IP calls

12:00 and to shut the media streams between the purchasers another important topic to discuss is quality of service qos in

12:09 viip qos is employed for mensuration of the degree of user satisfactions in
12:16 different words quality of service will be outlined because the network ability to
produce smart services that satisfy
12:24 its customers once degree of user satisfaction is over it mean that Qs is additionally
12:32 higher Qs area unit shortly represented as given below delay delay will be outlined
because the
12:41 total time it takes since an individual act another person speak words and
12:48 hearing them at the opposite finish delay will be classified into three categories
delay at the supply delay at
12:57 the receiver and and network delay Jitter variation in transmission
13:05 delay is thought as interference science network doesn't guarantee of package
delivery time that introduces variations
13:13 in transmission delay and its additional negative effects on voice quality packet
laws packets transmitted over science
13:22 network is also lost within the network or arrived corrupted or late packets would
be discarded guarded after they
13:30 arrive late at the interference buffer of the receiver or once there's overflow in the
interference buffer or router
13:36 buffer therefore packet loss is adequate the full loss happens throughout
congestion of network and late
13:44 arrival throughout the packet loss the sender is aware to conduct the Lost packets
it causes additional packet
13:52 delay and it affects the transmission Qs Eco in VII
13:59 Eco happens once a caller at the center aspect hears the reflection of his own voice
once he talked on the phone or the
14:09 electroacoustic transducer whereas the coli doesn't notice the Eco Eco is that
14:16 the term of the reflections of the scent voice signal by the way finish Eco May will
be electrical Eco that exists in
14:25 pstn networks or Eco of sound that is difficulty in viip networks
14:33 throughput the output is also outlined because the most range of bits received out
of the full range of bits sent
14:42 throughout associate degree interval of your time next topic we are going to
discuss is viip
14:51 Security First is viip attacks threats attackers typically Target some
14:58 fashionable and wellknown systems and applications viip has become one in every
of such
15:05 application viip like every another system or applications have its weakness
therefore protocol designers ought to
15:13 address it before with the success putting in viip on a universal scale thus this

section presents a study of

15:22 attacks on the viip Dos denial of service dos attacks that reduces the
15:30 quantity of obtainable science addresses information mayor and different router
15:36 functions a DS attack typically blocks the service of the server a viip based
15:43 mostly DS attack bombards a decision process application with the massive
amounts of coincidental requests that it
15:51 cannot method inflicting the motility down of the appliance thereby denying service
to approved or supposed
16:00 users DS attacks will be directed toward any network part to disrupt these systems
16:08 practicality Network sniffing Network sniffing attacks occur
16:15 once a personal or offender is observant the network traffic patterns typically
16:22 any system user or attacker on a network that's sharing a transmission medium has
the power how to look at different
16:30 systems traffic eavesdropping eavesdropping is a shot towards assembling
sensitive data to
16:38 arrange for a Cyber attack or to achieve intelligence in viip E dropping maybe a
16:45 situation wherever the offender is in a position to watch signal or media contents
that area unit changed between
16:53 users so as to look at communications to arrange for different future attack ta
17:01 17 minutes, 1 second spoofing one of the kinds of viip spoofing is display spoofing
wherever the offender
17:08 masqueraders itself as a licensed viip user and places a decision the display that
seems on the
17:16 user aspect seems authentic even supposing the offender is exploitation unfair
means that the offender will
17:25 currently trick the user in giving for free sensitive information spoofing is thus the
VIP
17:32 version of ancient fishing toall fraud tall fraud is that
17:39 the ability to possess associate degree unauthorized access to the VIP Services
typically for financial gain for VIP
17:48 suppliers this is often one in every of the for most essential attacks toall FR
17:56 will be recognized by the manipulating the sign messages or the configuration of
VIP Parts together with the charge
18:04 systems spam over net telephone that is spit due to its a lot of lower
18:11 communication prices VIP network has become additional engaging as another to
18:17 this bstn further as a target for spammers viip spam may be a recorded
18:25 self- dialed phone calls exploitation foration net protocol VIP spam
18:32 additionally called spam over net telephone that is spit is popping Dead Set be a
significant downside for VIP

18:42 networks spit is additional sever that your email spam due of its assaultive nature which needs a period defense

18:51 reaction what is the security measures for viip according downside on DUS there has been a report that sure

19:00 viip phones area unit Lael to each do attacks and sure viip routers also are

19:08 at risk of malicious Traffic Solutions to avoid a DS attack

19:15 monitoring and filtering to take care of list of suspicious users and deny those users from getting any connection or

19:24 session authentication to ATT test the identity of a US user before forwarding his or her messages through the

19:33 network stateless proxy to lower the risks of memory exhaustion du as

19:40 homeless proxy will be accustomed to perform different Security checks like user authentication third party

19:48 registration and filtering spam sources server style to create computer

19:55 hardware memory and network affiliation the primary line of defense against any dos

20:03 attacks reported downside on EES dropping an Internet Security Systems

20:10 team discovered the issues in viip system during a vendor decision manager the vendors offer associate

20:19 degree offender the power to concentrate in or forward calls additionally to gain unauthorized access to networks running

20:27 on VI answer to eavesdropping employing perfect

20:34 Hardware only approved forx or to tend access to wiring

20:41 closets vulnerable Network purpose or to implement a port based mostly waterproof

20:48 address security for instance on a reception ctsy phone a method ought to often scan the

20:57 network of devices is running in unauthorized mode another answer is secret writing of

21:05 viip traffic reported downside on spoofing there are many reports that Banks and

21:13 online payment services were victims of attack wherever the offender referred to as a credit card client and receiv

21:21 received the purchases into giving for free sensitive data relating to their accounts by declaring that there had

21:28 been dishonorable activity on their accounts solution to spoofing an effective authentication module combined

21:37 with secret writing would be an efficient answer to spoofing and masquerading

21:44 attacks reported downside on tall fraud tall fraud threat is commonly most

21:51 vital within the eyes of enormous organization an eminent theory of service from

from an outsized

21:59 organization will go unnoticed for quite your time permitting the offender to rack up an outsized bill at the

22:08 organization's expense but tall fraud is additionally quite simple to accomplish a straightforward configuration downside

22:18 within the dial set up with leave lines open for check purpose associate degreed remote exess will be abused by an

22:26 offender it t but profitable from the purpose of fre of the offender to commit tall fraud as a result of once the

22:36 attack is over the offender will abuse the service quite speedily by reselling them with no real prices concerned from

22:45 the attackers aspect solution to tall fraud viip suppliers will stop tall

22:53 fraud by configuring powerful Fireballs that stop attacks and by protective the

23:01 23 minutes, 1 second ports reported downside on spit it is potential Network for an offender to

23:09 impersonate as another VIP caller for instance associate degree offender May

23:15 probably inject a pretend ID into a standard VIP decision so the receiver mistakes the decision to be real and

23:24 coming back from a best known and short supply the receiver mistaken by the electronic

23:32 ID of the caller might Place unwarranted trust within the person at the opposite finish in such a communication the

23:41 receiver is also hawed into revealing personal data like account number SSN or

23:48 answers to security questions a mother's family name for instance solution to spit a normal blacklisting approach l

23:58 through all the calls coming back from the best known Science address VIP vendors additionally give security

24:06 measurements to Dam suspicious callers above all receiver ought to be allowed to not break any vital data to

24:15 the offender next topic is configuration of viip adapters USB viip phone adapters

24:25 enable USA to use any as ient telephone to position VIP calls they typically

24:33 appear as if USB adapters these USB adapters have a typical standard Telephone Phone Port so

24:42 that one will connect a daily connective once hooked up your phone operates as if connected to traditional

24:51 public utility software controlled VIP applications soft phones soft phone's area unit phones

25:00 that enable USA to create viip phone calls directly from a PC that has a web

25:07 affiliation soft phones build decision with the assistance of a laptop telephone

receiver and a sound card a

25:15 soft phone sort of a traditional phone just with the distinction that the affiliation is coming back from your

25:23 laptop viip suppliers devel soft phones for free of charge in exchange of availing their

25:30 service code controlled VIP application permits users to speak to others dedicated VIP phones

25:40 exploitation these services at no further price a viip phone feels like a

25:47 daily telephone an evid viip phone connects on an electronic network instead of a standard telephone line an

25:56 evid VIP phone May match in two ways that a phone and a base station that

26:04 connects to the net it should operate on a neighborhood wireless network dedicated viip phones need a

26:13 supplier further as a service setup dedicated routers these routers in viip enable

26:22 user to attach standard phones to the net to position VIP calls wherever the

26:29 router is connected to associate degree ADSL cable electronic equipment and

26:36 permit users to connect a standard telephone viip routers have the extra practicality of associate degree science

26:45 router as that permits USA to attach to the laptop further viip suppliers piece

26:53 these routers underneath a particular setup for a charge these routers may also perform their functions freelance

27:01 27 minutes, 1 second of any PC or any code Wireless compatibility of

27:07 viip with the wireless router mobile devices and smartphones may also use the

27:14 viip system if you put in a wireless computer network you would like a applicable security like a firewall or

27:23 secret writing problems with wireless VIP VIP on computer network is deployed

27:31 principally in company environments like in corporations instead of homes Wireless VIP poses issues of

27:40 measurability for Enterprises as just in case with all wireless network quality of service is

27:48 poor compared to wide networks setup price and maintain prices higher during a wireless VIP

27:56 Network due to variety of exess points among the restricted space of network security threat is higher over a

28:05 wireless VIP Network limitations of viip the popularity of viip can rely

28:13 upon some key problems a number of these problems area unit a resultant of the very fact that science was designed for

28:22 information packets whereas some problems stem from vendors that don't seem to be Mee meeting the Necessities to the

28:30 standards the key problems area unit mentioned as follows quality of service the
design of science doesn't guarantee

28:39 realtime transmission of voice packets as science was designed for information
packets that guarantee error-free

28:48 ordered delivery of information packets acceptableness of viip depends

28:55 on delay that should not exceed a given threshold worth prioritization of packets
means

29:03 that giving voice packets high priority will guarantee smart quality of voice

29:11 interoperability you need to Interchange the sign mechanism of pstn with viip sign
mechanism if you wish to create

29:19 viip common among net users a number of the suitable sign mechanisms area unit

29:27 h. 3 to three standards sip protocols and mgcp every of those protocols integrates

29:36 information voice and video over constant wire security the question of security
arises as a result of the

29:44 backbone of uip is that the net that isn't a really secure medium there will

29:51 be interception of calls due as fraud Security will be provided by exploit ation
tunneling protocols like

30:00 layer to tunneling the secret writing mechanism used in Secure sockets layer

30:07 that is SSL however secret writing isn't wide out there for viip integration with

30:14 public switch telephone Network that is pstn viip Works in Union with pstn and

30:22 that they seem as one network to the users of the service in viip technology your
telephone number has associate

30:31 degree Science address therefore anytime a viip phone engages during a decision

30:38 its Science address is translated into the telephone number and two-handed over to
the pstn network you would like to

30:48 mix of each viip a pstn as a result of not everybody has switched to viip and

30:55 their area unit are closer range of users still exploitation pstn

31:03 scalability since calls over science have lower price and also the work on the rising
the standard of voice further

31:11 as transmission goes on there has been a high rate in VIP users the most obstacle
lies in its

31:19 measurability so viewers in today's lecture we discussed viip Protocols of viip Qs in
viip and and security

31:28 implementation on viip we also discussed the security measur and configuration of

31:35 viip we will discuss the VIP protocols in detail in forthcoming lectures thanks

31:41 for watching this program thank you